

# SQL Fundamentals

## Updated Exercises

### Exercise 4 : SQL Date Functions

#### Questions 1-5 - Extract Date Parts

##### Q1. Day Name from Order Date

```
SELECT order_id,  
       customer_id,  
       order_date,  
       DAYNAME(order_date) AS day_name  
FROM orders;
```



| order | customer_id | order date | day_name |
|-------|-------------|------------|----------|
| 1001  | 101         | 2026-05-01 | Friday   |
| 1002  | 102         | 2026-05-02 | Saturday |
| 1003  | 103         | 2026-05-03 | Sunday   |
| 1004  | 104         | 2026-05-04 | Monday   |
| 1005  | 105         | 2026-05-05 | Tuesday  |

## Q2. Month Name from Signup Date

```
SELECT customer_id,  
       customer_name,  
       signup_date,  
       MONTHNAME(signup_date) AS signup_month_name  
FROM   customer_signups;
```

↓

| customer_id | customer_name | signup_date | signup_month_name |
|-------------|---------------|-------------|-------------------|
| 201         | John          | 2026-01-15  | January           |
| 202         | Mary          | 2026-02-20  | February          |
| 203         | Peter         | 2026-03-05  | March             |
| 204         | Sarah         | 2026-04-18  | April             |
| 205         | Thabo         | 2026-05-30  | May               |

## Q3. Month Number from Sales Date

```
SELECT sale_id,  
       product_name,  
       sale_date,  
       MONTH(sale_date) AS sale_month  
FROM   sales;
```

↓

| sale_id | product name | sale_date  | sale_month |
|---------|--------------|------------|------------|
| 5001    | Laptop       | 2026-01-10 | 1          |
| 5002    | Mouse        | 2026-02-15 | 2          |
| 5003    | Keyboard     | 2026-03-20 | 3          |
| 5004    | Monitor      | 2026-04-25 | 4          |
| 5005    | Desk         | 2026-05-30 | 5          |

#### Q4. Year from Transaction Date

```
SELECT transaction_id,  
       customer_id,  
       transaction_date,  
       YEAR(transaction_date) AS transaction_year  
FROM transactions;
```



| transaction_id | customer_id | transaction_date | transaction_year |
|----------------|-------------|------------------|------------------|
| T001           | 101         | 2024-12-15       | 2024             |
| T002           | 102         | 2025-01-20       | 2025             |
| T003           | 103         | 2025-06-10       | 2025             |
| T004           | 104         | 2026-07-05       | 2026             |
| T005           | 105         | 2026-05-25       | 2026             |

#### Q5. Day of Month from Delivery Date

```
SELECT delivery_id,  
       customer_id,  
       delivery_date,  
       DAY(delivery_date) AS day_of_month  
FROM deliveries;
```



| delivery_id | customer_id | delivery_date | day_of_month |
|-------------|-------------|---------------|--------------|
| D001        | 101         | 2026-05-01    | 1            |
| D002        | 102         | 2026-05-08    | 8            |
| D003        | 103         | 2026-05-15    | 15           |
| D004        | 104         | 2026-05-22    | 22           |
| D005        | 105         | 2026-05-29    | 29           |

## Questions 6-10 - Current Date, Convert & Format

### Q6. Add Today's Date

```
SELECT employee_id,  
       employee_name,  
       department,  
       CURRENT_DATE() AS today_date  
FROM employees;
```

→ 20-07-2026

| employee_id | employee_name | department | today_date |
|-------------|---------------|------------|------------|
| 301         | Nandi         | Sales      | 2026-07-20 |
| 302         | Brian         | IT         | 2026-07-20 |
| 303         | Lerato        | Finance    | 2026-07-20 |
| 304         | Sipho         | HR         | 2026-07-20 |
| 305         | Asha          | Marketing  | 2026-07-20 |

### Q7. Convert Text to Date

(works if date column is a time stamp)

```
SELECT order_id,  
       customer_id,  
       order_date_text,  
       TO_DATE(order_date_text, 'YYYY-MM-DD') AS order_date  
FROM online_orders;
```

| order_id | customer_id | order_date_text | order_date |
|----------|-------------|-----------------|------------|
| 4001     | 101         | 2026-01-15      | 2026-01-15 |
| 4002     | 102         | 2026-02-20      | 2026-02-20 |
| 4003     | 103         | 2026-03-25      | 2026-03-25 |
| 4004     | 104         | 2026-04-10      | 2026-04-10 |
| 4005     | 105         | 2026-05-05      | 2026-05-05 |

### Q8. Format Date as Text

```
SELECT payment_id,  
       customer_id,  
       payment_date,  
       TO_CHAR(payment_date, 'YYYY-MM-DD') AS formatted_payment_date  
FROM payment_dates;
```

| payment_id | customer_id | payment_date | formatted_payment_date |
|------------|-------------|--------------|------------------------|
| P001       | 101         | 2026-01-05   | 2026-01-05             |
| P002       | 102         | 2026-02-10   | 2026-02-10             |
| P003       | 103         | 2026-03-15   | 2026-03-15             |
| P004       | 104         | 2026-04-20   | 2026-04-20             |
| P005       | 105         | 2026-05-25   | 2026-05-25             |

### Q9. Days Since Last Purchase

```
SELECT customer_id,  
       customer_name,  
       last_purchase_date, → 2026-07-20  
       DATEDIFF(CURRENT_DATE(), last_purchase_date) AS  
       days_since_last_purchase  
FROM customer_purchases;
```

| customer_id | customer_name | last_purchase_date | day_since_last_purchase |
|-------------|---------------|--------------------|-------------------------|
| 501         | John          | 2026-05-01         | 80                      |
| 502         | Mary          | 2026-05-10         | 71                      |
| 503         | Peter         | 2026-05-15         | 66                      |
| 504         | Sarah         | 2026-05-20         | 61                      |
| 505         | Thabo         | 2026-05-25         | 56                      |

Q.10. Expected Delivery Date → order\_date + 7 days a normal order usually takes

```
SELECT order_id,
       customer_id,
       order_date,
       DATEADD(DAY, 7, order_date) AS expected_delivery_date
FROM shipping-orders;
```

| order_id | customer_id | order_date | expected_delivery_date |
|----------|-------------|------------|------------------------|
| 6001     | 101         | 2026-05-01 | 2026-05-08             |
| 6002     | 102         | 2026-05-03 | 2026-05-10             |
| 6003     | 103         | 2026-05-05 | 2026-05-12             |
| 6004     | 104         | 2026-05-07 | 2026-05-14             |
| 6005     | 105         | 2026-05-09 | 2026-05-16             |

### Question 11-15 - Multi-Extract & Date Filtering

Q.11. Year, Month, Day Separately

```
SELECT booking_id,
       customer_id,
       booking_date,
       YEAR(booking_date) AS booking_year,
       MONTH(booking_date) AS booking_month,
       DAY(booking_date) AS booking_day
FROM bookings;
```

| booking_id | customer_id | booking_date | booking_year | booking_month | booking_day |
|------------|-------------|--------------|--------------|---------------|-------------|
| B001       | 101         | 2026-01-12   | 2026         | 1             | 12          |
| B002       | 102         | 2026-02-18   | 2026         | 2             | 18          |
| B003       | 103         | 2026-03-22   | 2026         | 3             | 22          |
| B004       | 104         | 2026-04-09   | 2026         | 4             | 09          |
| B005       | 105         | 2026-05-27   | 2026         | 5             | 27          |

Q12. Filter Orders from 2026

```
SELECT order_id,  
       customer_id,  
       order_date,  
       YEAR(order_date) AS order_year,  
       amount  
FROM yearly_orders  
WHERE YEAR(order_date) = 2026;
```



| order_id | customer_id | order_date | order_year | amount |
|----------|-------------|------------|------------|--------|
| 7004     | 104         | 2026-02-05 | 2026       | 1500   |
| 7005     | 105         | 2026-05-25 | 2026       | 2000   |

Q13. Filter Orders from March

```
SELECT order_id,  
       customer_id,  
       order_date,  
       MONTH(order_date) AS order_month,  
       amount  
FROM monthly_orders  
WHERE MONTH(order_date) = 3;
```



| order_id | customer_id | order_date | order_month | amount |
|----------|-------------|------------|-------------|--------|
| 8003     | 103         | 2026-03-10 | 3           | 800    |
| 8004     | 104         | 2026-03-25 | 3           | 1500   |

## Q14. Last Day of the Month

```
SELECT subscription_id,  
       customer_id,  
       start_date,  
       LAST_DAY(start_date) AS month_end_date  
FROM   subscriptions;
```



| Subscription_id | customer_id | start_date | month_end_date |
|-----------------|-------------|------------|----------------|
| SUB001          | 101         | 2026-01-10 | 2026-01-31     |
| SUB002          | 102         | 2026-02-15 | 2026-02-28     |
| SUB003          | 103         | 2026-03-20 | 2026-03-31     |
| SUB004          | 104         | 2026-04-25 | 2026-04-30     |
| SUB005          | 105         | 2026-05-30 | 2026-05-31     |

## Q15. First Day of the Month

```
SELECT send_id,  
       customer_id,  
       send_date,  
       DATE_TRUNC('MONTH', send_date) AS month_start_date  
FROM   campaign_sends;
```



| send_id | customer_id | send_date  | month_start_date |
|---------|-------------|------------|------------------|
| C001    | 101         | 2026-01-12 | 2026-01-01       |
| C002    | 102         | 2026-02-18 | 2026-02-01       |
| C003    | 103         | 2026-03-22 | 2026-03-01       |
| C004    | 104         | 2026-04-09 | 2026-04-01       |
| C005    | 105         | 2026-05-27 | 2026-05-01       |

## Question 16-20 - Advanced Date Work

### Q16. Format as Month and Year

```
SELECT invoice_id,  
       customer_id,  
       invoice_date,  
       TO_CHAR (invoice_date, 'MMMM YYYY') AS invoice_month_year  
FROM invoice_dates;
```

↓

| invoice_id<br><del>customer</del> | customer_id | invoice_date | invoice_month_year |
|-----------------------------------|-------------|--------------|--------------------|
| INV001                            | 101         | 2026-01-05   | January 2026       |
| INV002                            | 102         | 2026-02-10   | February 2026      |
| INV003                            | 103         | 2026-03-15   | March 2026         |
| INV004                            | 104         | 2026-04-20   | April 2026         |
| INV005                            | 105         | 2026-05-25   | May 2026           |

### Q17. Customer Age from Date of Birth

```
SELECT customer_id,  
       customer_name,  
       date_of_birth,  
       FLOOR (DATEDIFF (CURRENT_DATE(), date_of_birth) / 365.25) AS  
       customer_age  
FROM customers_birthdays;
```

↓

| customer_id | customer_name | date_of_birth | customer_age |
|-------------|---------------|---------------|--------------|
| 901         | John          | 1998-05-10    | 28           |
| 902         | Mary          | 1990-08-20    | 35           |
| 903         | Peter         | 2002-03-15    | 24           |
| 904         | Sarah         | 1985-12-01    | 40           |
| 905         | Thabo         | 2000-07-30    | 25           |

Q18. Classify Weekend vs Weekday

```
SELECT order_id,  
       customer_id,  
       order_date,  
       DAYNAME(order_date) AS day_name,  
       CASE  
           WHEN DAYNAME(order_date) IN ('Saturday', 'Sunday')  
           THEN 'Weekend'  
           ELSE 'Weekday'  
       END AS day_type  
FROM weekend_orders;
```

↓

| order_id | customer_id | order_date | day_name | day_type |
|----------|-------------|------------|----------|----------|
| 9001     | 101         | 2026-05-01 | Friday   | Weekday  |
| 9002     | 102         | 2026-05-02 | Saturday | Weekend  |
| 9003     | 103         | 2026-05-03 | Sunday   | Weekend  |
| 9004     | 104         | 2026-05-04 | Monday   | Weekday  |
| 9005     | 105         | 2026-05-05 | Tuesday  | Weekday  |

### Q19. Transactions by Quarter

```
SELECT transaction_id,  
       customer_id,  
       transaction_date,  
       QUARTER(transaction_date) AS transaction-quarter,  
       amount
```

```
FROM quarterly_transactions ;
```

| transaction_id | customer_id | transaction_date | transaction-quarter | amount |
|----------------|-------------|------------------|---------------------|--------|
| Q001           | 101         | 2026-01-15       | 1                   | 500    |
| Q002           | 102         | 2026-03-20       | 1                   | 1200   |
| Q003           | 103         | 2026-04-10       | 2                   | 800    |
| Q004           | 104         | 2026-07-05       | 3                   | 1500   |
| Q005           | 105         | 2026-10-25       | 4                   | 2000   |

### Q20. Orders older than 30 days

```
SELECT order_id,  
       customer_id,  
       order_date,  
       DATEDIFF(CURRENT_DATE(), order_date) AS days-since-order,  
       amount
```

```
FROM recent_orders
```

```
WHERE DATEDIFF(CURRENT_DATE(), order_date) > 30 ;
```

| order_id | customer_id | order_date | days-since-order | amount |
|----------|-------------|------------|------------------|--------|
| R001     | 101         | 2026-04-01 | 110              | 500    |
| R002     | 102         | 2026-04-15 | 96               | 1200   |
| R003     | 103         | 2026-05-01 | 80               | 800    |
| R004     | 104         | 2026-05-10 | 71               | 1500   |
| R005     | 105         | 2026-05-25 | 56               | 2000   |

## Bonus Challenge

### Customer Recency Segment

```
SELECT customer_id,  
       customer_name,  
       last_purchase_date  
       DATEDIFF(CURRENT_DATE(), last_purchase_date) AS days_since_last_purchase  
CASE  
    WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date) <= 30  
    THEN 'Active Customer'  
    WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date)  
    BETWEEN 31 AND 90  
    THEN 'At Risk Customer'  
    ELSE 'Inactive Customer'  
END AS customer_status  
FROM customer_recency;
```



| customer_id | customer_name | last_purchase_date | days_since_last_purchase | customer_status   |
|-------------|---------------|--------------------|--------------------------|-------------------|
| 1001        | John          | 2026-05-25         | 56                       | At Risk customer  |
| 1002        | Mary          | 2026-05-10         | 71                       | At Risk customer  |
| 1003        | Peter         | 2026-04-01         | 110                      | Inactive customer |
| 1004        | Sarah         | 2026-02-15         | 155                      | Inactive customer |
| 1005        | Thaloo        | 2025-12-20         | 212                      | Inactive customer |